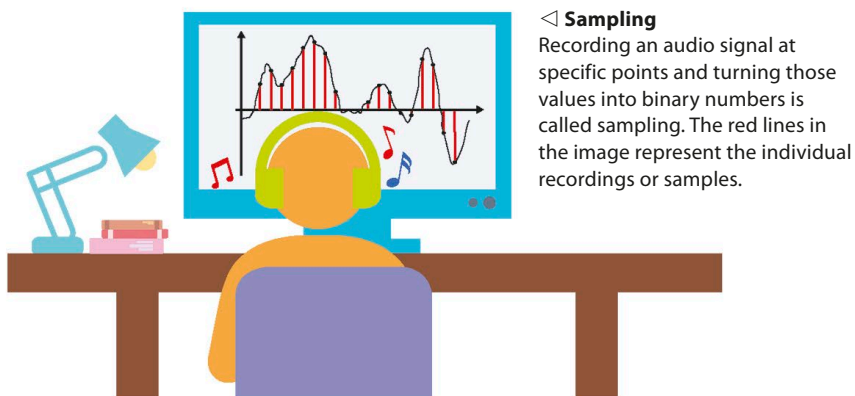


Encoding audio and video

Technology has transformed the way we consume and play audio and video. Digital music has completely overtaken physical storage systems, while digital video has replaced older analogue systems.

Encoding audio files

To encode an audio signal, its value is recorded thousands of times a second. Each of these values is turned into a binary number, which encodes not just the pitch of the note but other information, such as how loud it is. Playing back the sample values in the same order they were recorded produces a sound that, to the human ear, is identical to the original.



◀ Sampling

Recording an audio signal at specific points and turning those values into binary numbers is called sampling. The red lines in the image represent the individual recordings or samples.

SEE ALSO

◀ 16–17 Computing for you

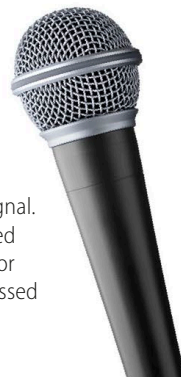
◀ 80–81 Bits and digitization

◀ 82–83 Binary code

REAL WORLD

How microphones work

Sounds are waves that move through the air by compressing it at regular intervals. Microphones contain a thin piece of metal or plastic called a diaphragm, which vibrates when it is hit by sound waves. Electronics in the microphone translate the change in vibration into an electrical signal. This can be amplified through a speaker, or digitized and processed by a computer.



What affects audio quality?

In audio encoding, the number of samples taken per second is called the sample rate. The more samples taken per second, the more accurately the sound is represented. The number of bits used to store each sample is known as the bit depth. A higher bit depth means more information can be stored, which leads to better quality recordings. Combining the sample rate and bit depth gives the bit rate, or bits used per second.

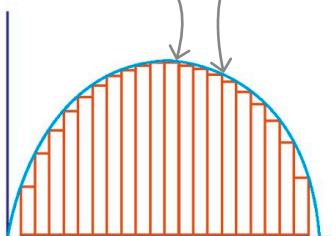
▷ High values lead to better sound

Each rectangle in these graphs represents a sample. With a higher sample rate, the rectangles are narrower and fit together to form a shape that is closer to the waveform's shape.

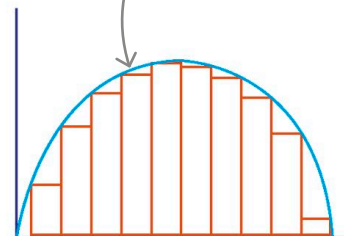
The waveform is the shape and form of the musical signal.

High sample rate allows a better fit to the waveform.

Low sample rate fits the waveform less accurately, and more information is lost.



High sample rate



Low sample rate